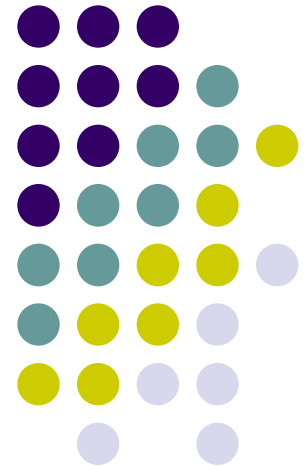
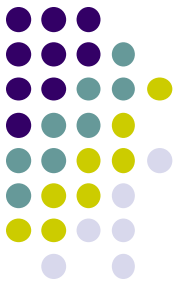


Integrating Domain Models within Agent Models

Dynamic Modelling for
Human-centered Systems

Chapter 14

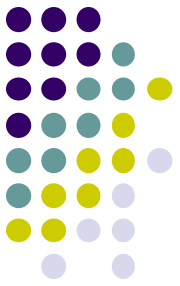




Summary previous lecture

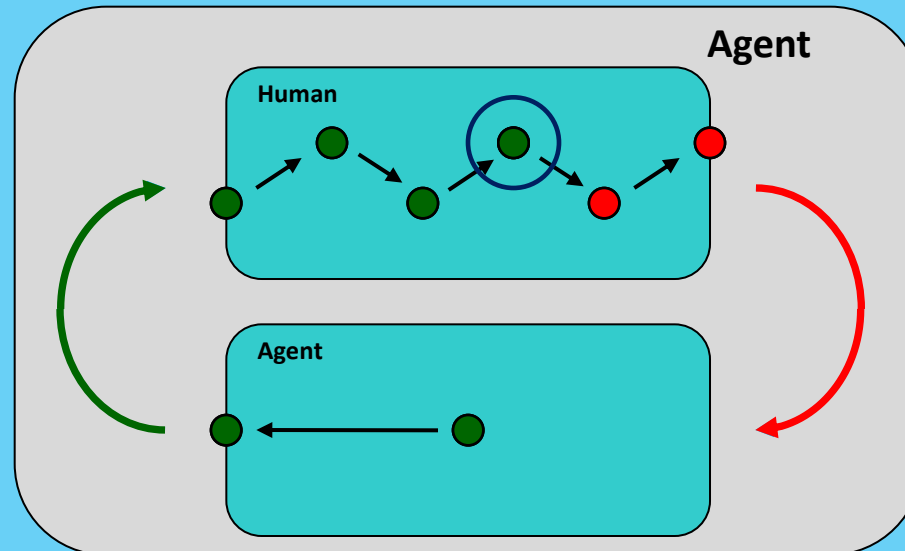
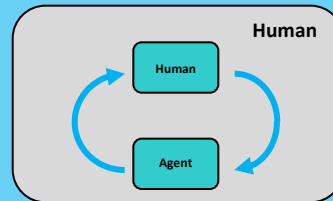
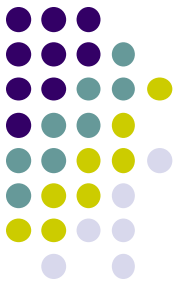
- **Agent:**
 - Capable of autonomous action in environment to achieve goals
- **Intelligent Agent:**
 - Reactive
 - Pro-active
 - Social
- **Basic agent concepts:**
 - Interface: observations, actions, communications
 - Internal: beliefs, desires, intentions

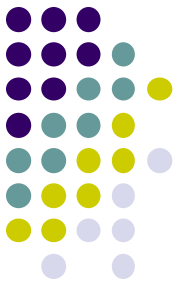
Integrating models in intelligent applications



By incorporating a model within an application, a smart system can be built that has **knowledge** about human functioning, and is able to **reason** about this

Global idea of integrating models

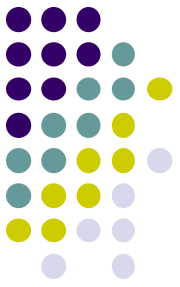




Two types of reasoning

- What is going on?
 - → **assessment** of situation
 - *analysis model*
- What will happen?
 - → **prediction** of consequence
 - *support model*

Overview of today's lecture



1. Integrating domain models within agent models

- analysis model
- support model
- adaptation model

2. Translating a domain model into an **analysis model**

- “ice cream” example of domain model
- translation into analysis model
- generic process

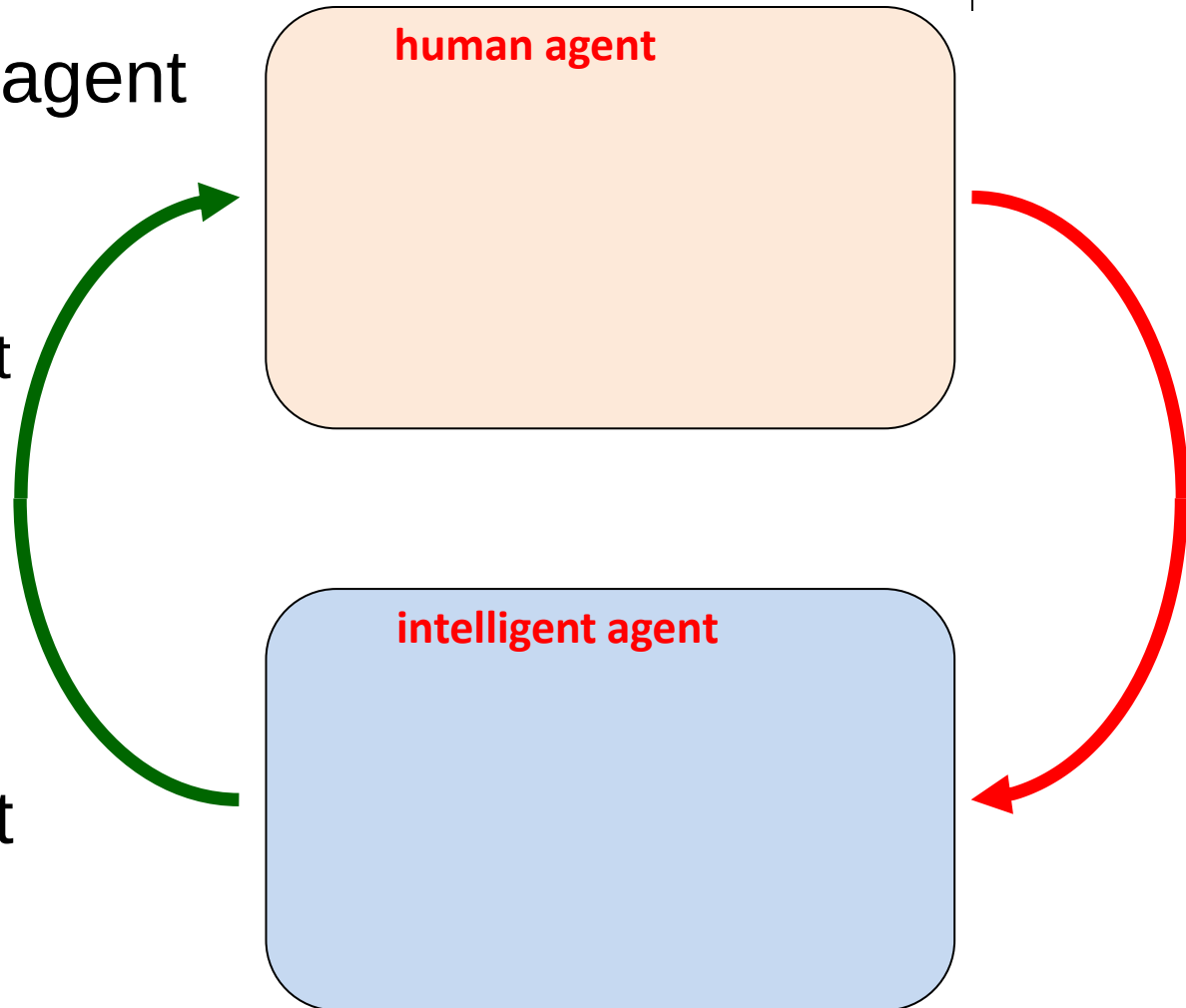
Next lecture: translating a domain model into a **support model**

Domain models and intelligent agents

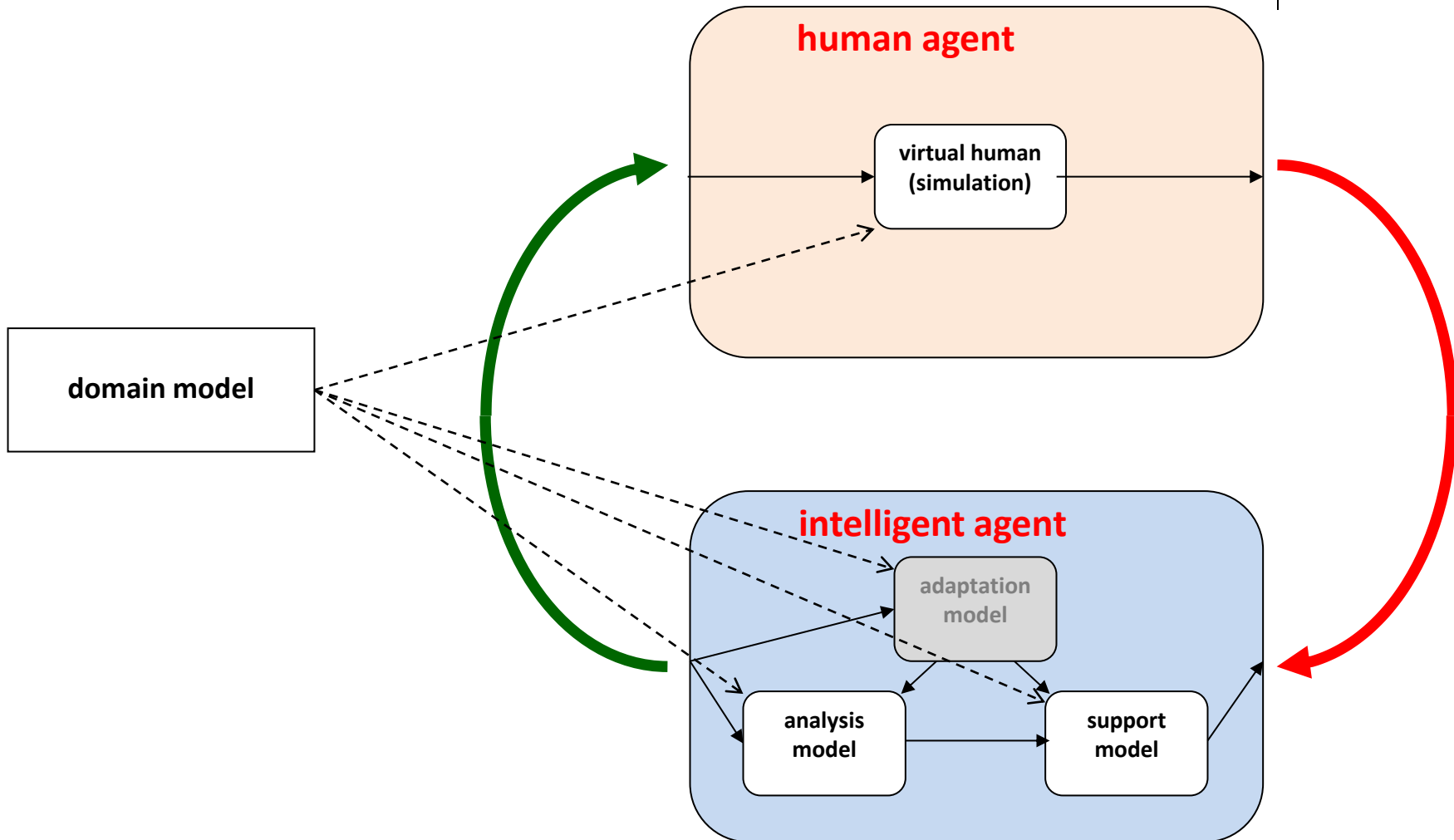
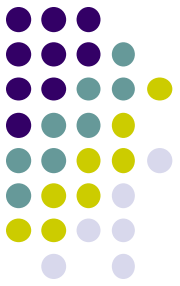


- Assumes multi-agent architecture
 - human is agent
 - intelligent agent

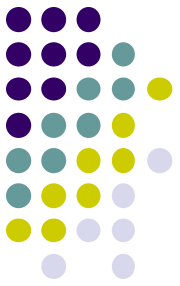
- Domain model forms basis of models used in intelligent agent



Four ways of use cases of domain model



Domain model integration: four ways



- **virtual human** model
 - directly represents human process
- **analysis** model
 - to analyse and assess the process
- **support** model
 - to generate support for the process
- **(parameter adaptation** model)
 - (to improve the model of the process)

Example: intelligent medicine box



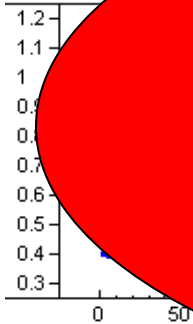
- Goal: remind people of medication intake based on medication level in blood
- What can be observed?
- Where are we interested in?





Example: medicine box

- Domain model can be used to simulate (virtual human):



IMPORTANT:

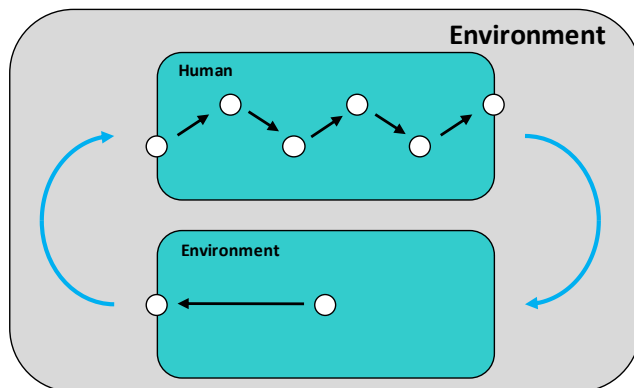
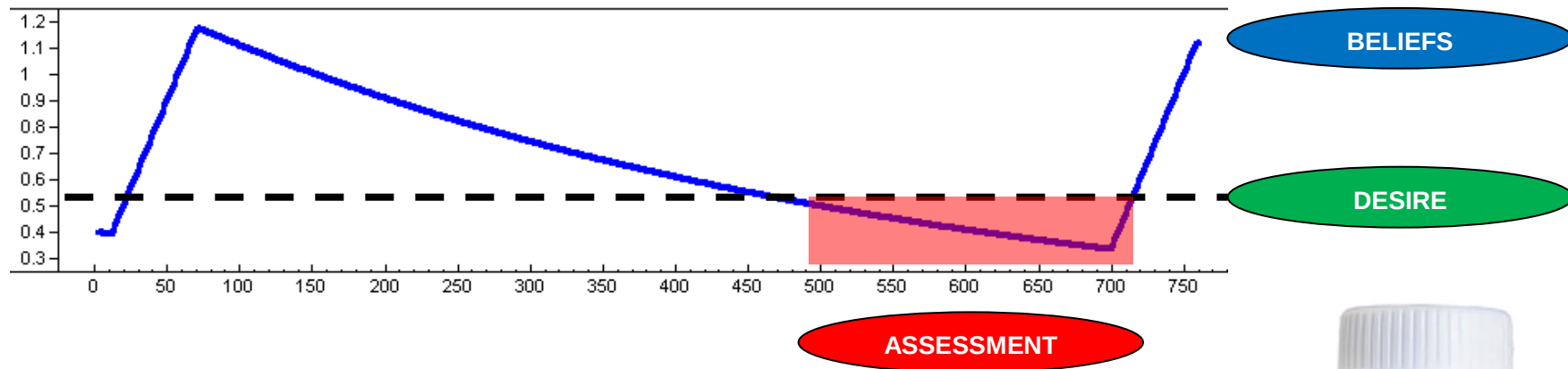
Domain Model $\neq$ Application!

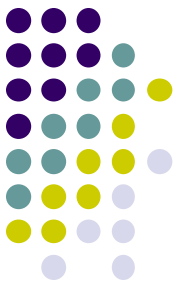




Example: medicine box

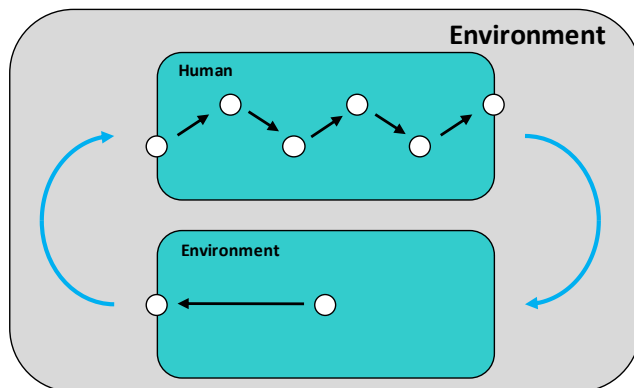
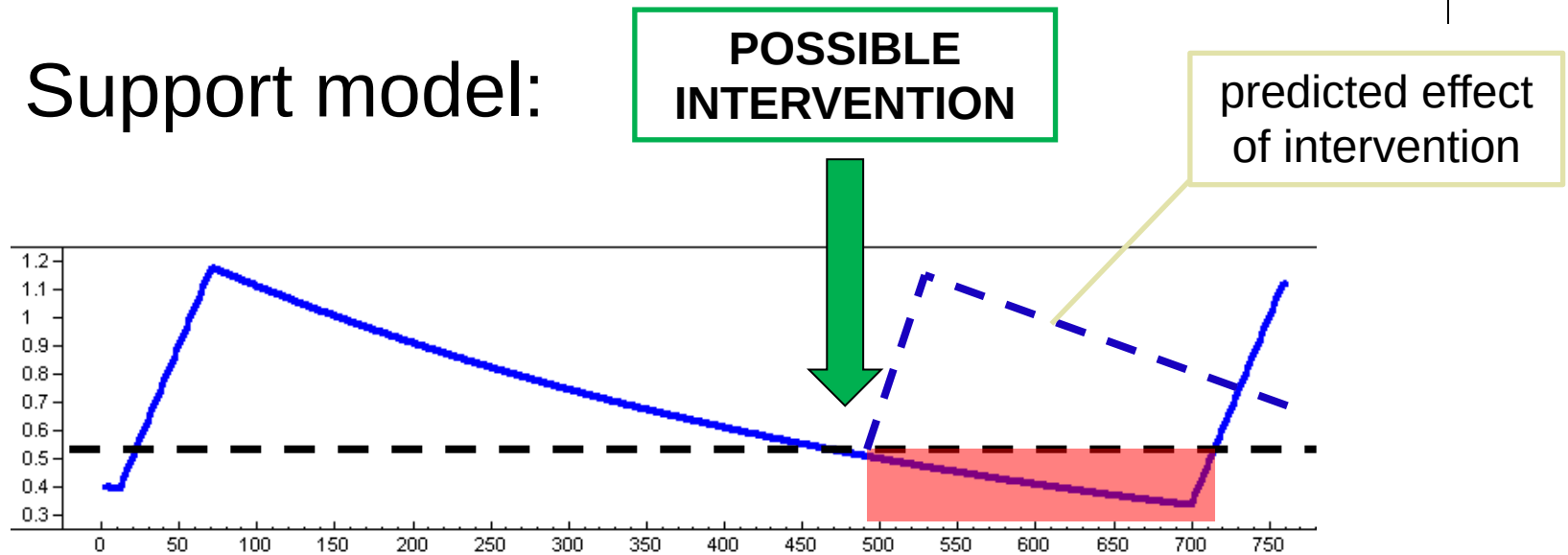
- Analysis model:





Example: medicine box

- Support model:





Main Steps - Summary

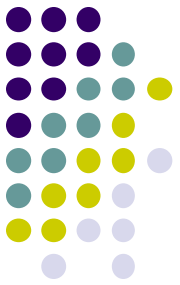
- **domain** model
 - formal description of human process
 - as developed during the first half of the course
- **analysis** model
 - uses domain model to interpret observations
 - matches *believed* human state to some *desired* state
 - ⇒ generates **assessments**
- **support** model
 - uses domain model to reason about effect of actions
 - ⇒ generates **support actions**

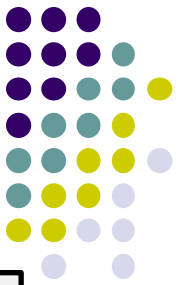
Question



*In addition to such models, **what else** is needed for intelligent applications?*

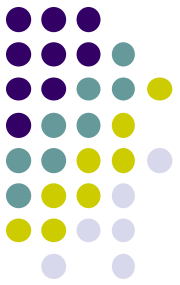
Data from sensors





Assignment

- Which aspects of human functioning can be measured using sensors?
- The next slide lists 12 aspects
- Name as many as you can



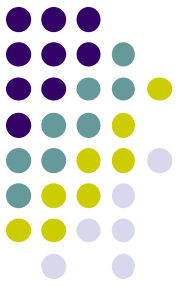
Aspects to be measured

- heart rate
- blood pressure
- skin conductance
- location
- movement
- facial expression
- eye movements
- voice
- temperature
- brain activity
- key strokes
- subjective experience
- ...



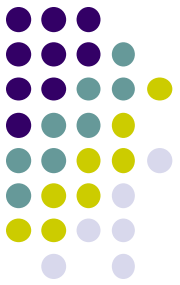
Overview of today's lecture

1. Integrating domain models within agent models
 - analysis model
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2. Translating a domain model into an analysis model
 - “ice cream” example of domain model
 - translation into analysis model
 - generic process



Example

- Modelling a simple behavior
→ **domain model**
- Using a translation of this model within an intelligent agent to asses some user state
→ **analysis model**

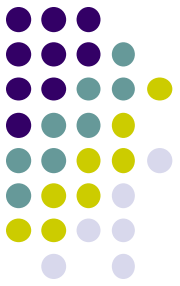


Ice cream domain model

- Behavior that we model:
 - Arnie crossing the street to buy an ice cream
- Factors that determine the behavior
 - the **observation** of an ice cream truck at the opposite of the street
 - the **desire** to buy an ice cream (because he likes it)
 - the state of being **hungry** (because not having eaten for some time)

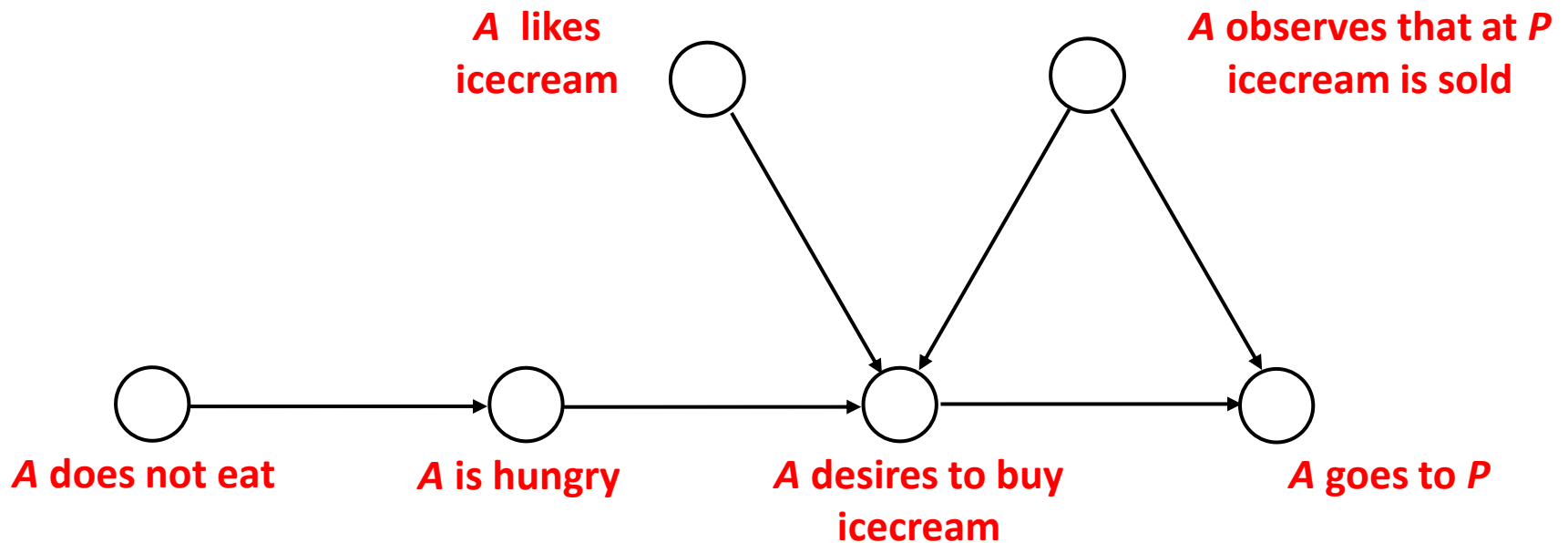
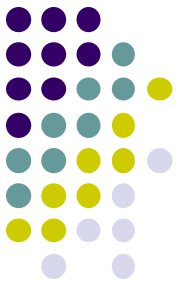


Main concepts used for the domain model

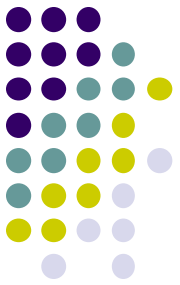


- *A* desires to buy icecream
- *A* observes that at *P* they sell icecream
- *A* goes to *P*
- *A* is hungry
- *A* does not eat
- *A* likes icecream

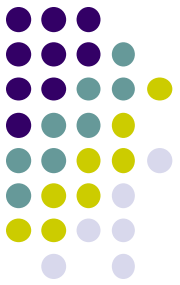
Dynamic relationships for the domain model



Detailed domain model: concepts



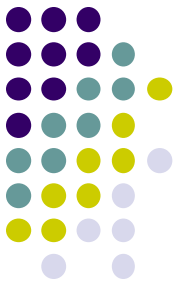
<i>concept</i>	<i>formalisation</i>
A desires to buy icecream	desire(A: AGENT, performed(A:AGENT, buy_icecream))
A observes that at <i>P</i> they sell icecream	observed(A: AGENT, icecream_at(P:POSITION))
A goes to <i>P</i>	performed(A: AGENT, goto(P:POSITION))
A is hungry	hungry(A: AGENT)
A likes icecream	likes(A: AGENT, icecream)
A does eat	performed(A: AGENT, eat)



Detailed domain model: sorts

<i>sort</i>	<i>description of use</i>	<i>elements</i>
AGENT	ambient and human agents	a, b, a1, a2, ..
ACTION	actions	eat, buy_icecream, goto(P:POSITION)
FOOD	types of food	icecream
POSITION	positions	p1, p2, ...
INFO_ELT	information element	icecream_at(P:POSITION)

Detailed domain model: formalization of relations



DDR1 Becoming hungry

If A does not eat
then A will be hungry

not performed(A, eat) ->> hungry(A)

DDR2 Generating the desire to buy icecream

If A is hungry
and A observes that at *P* they sell icecream
and A likes icecream
then A will desire to buy icecream

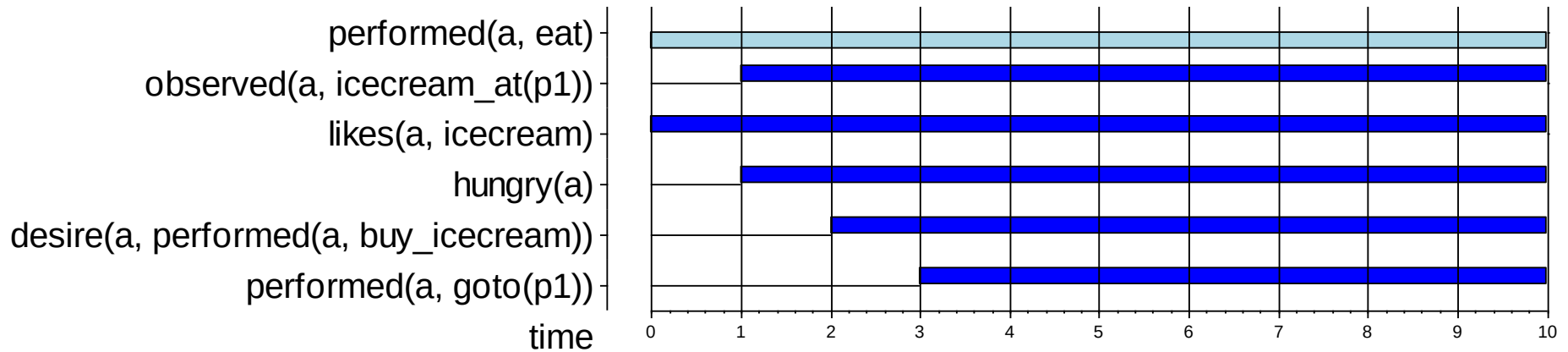
hungry(A) &
observed(A, icecream_at(P)) &
likes(A, icecream)
->> desire(A, performed(A, buy_icecream))

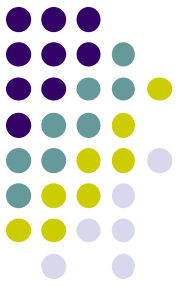
DDR3 Generating the action to go to the icecream

If A desires to buy icecream
and A observes that at *P* they sell icecream
then A will go to *P*

desire(A, performed(A:AGENT, buy_icecream)) &
observed(A, icecream_at(P))
->> performed(A, goto(P))

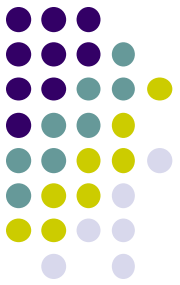
Example simulation trace for the domain model





Overview of today's lecture

1. Integrating domain models within agent models
 - analysis model
 - support model
 - adaptation model
2. Translating a domain model into an analysis model
 - “ice cream” example of domain model
 - **translation into analysis model**
 - generic process



Ice cream analysis model

- Idea: an **intelligent agent** that *reasons about* the behaviour of the **human agent** Arnie
- Goal of the system: finding out whether Arnie is hungry
(*Next lecture: help him solving this*)



Analysis by an intelligent agent



- Requires that **desire** is formulated
 - our example: Arnie should not be hungry
 - goal of the reasoning: assessing whether the desire is met or not
- Reasoning about observations can be done in two ways:
 - **forward in time** (from causes to effects)
 - **backward in time** (from effects to causes)



Agent concepts used

Interface agent concepts

- observations, incoming communications
- actions, outgoing communications

Internal agent concepts

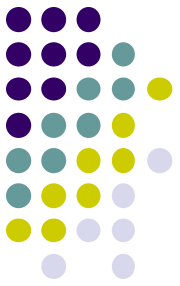
- beliefs, assumptions
- desires, liking, being hungry
- intentions
- assessments, predictions



How to formulate concepts?

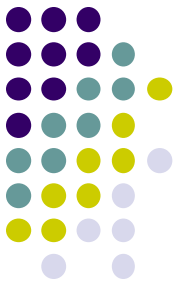
- First step: take perspective of intelligent agent
 - it **observes** Arnie and has **beliefs** about Arnie
- Then: add a goal (desire)
 - agent has a **desire** about Arnie
- Then: add conclusion

Concepts used for forward reasoning in an analysis model

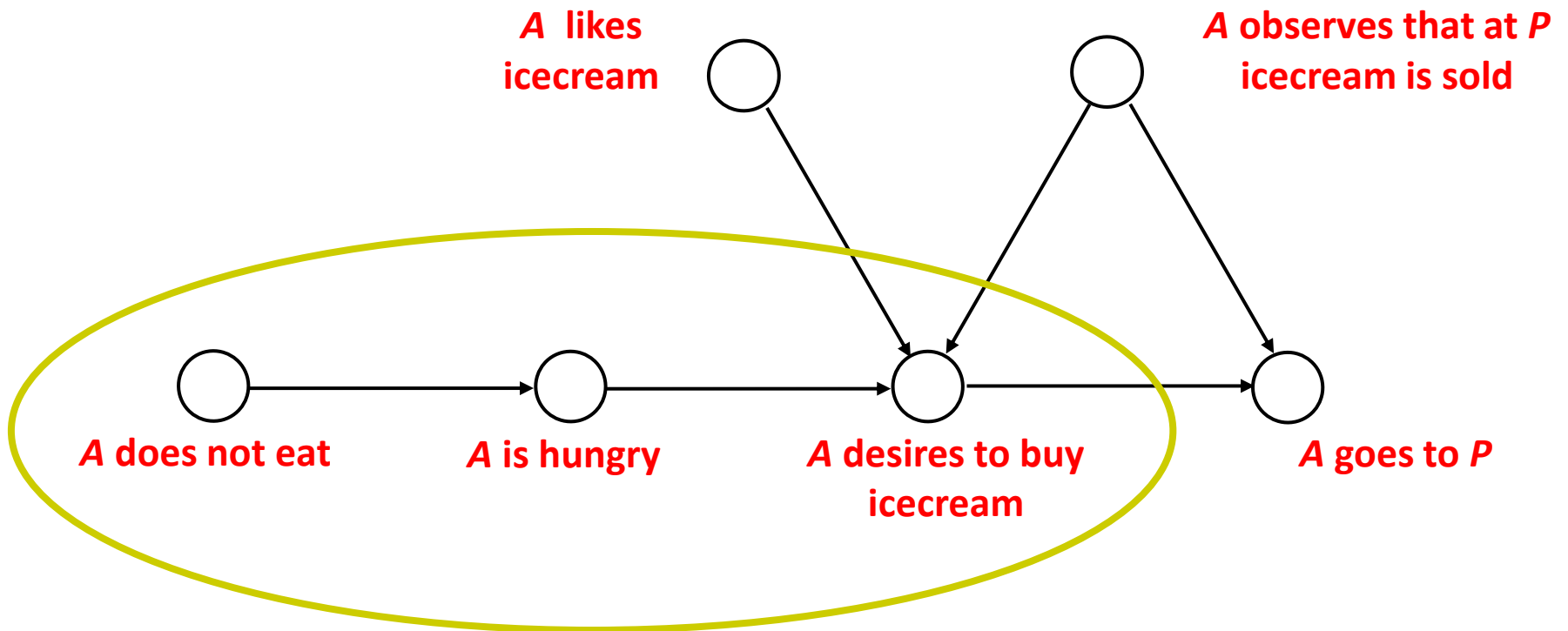
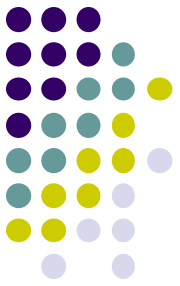


- ambient agent X observes Y
- ambient agent X believes Y
 - Y can be
 - A does not eat
 - A is hungry
 - A desires to buy icecream
 - A observes that at P they sell icecream
 - A likes icecream
 - A goes to P
- ambient agent X desires that A is not hungry
- ambient agent X assesses the undesirable fact that A is hungry

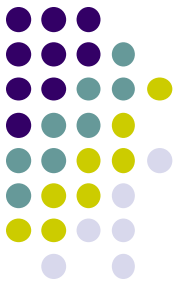
Analysis model via forward reasoning



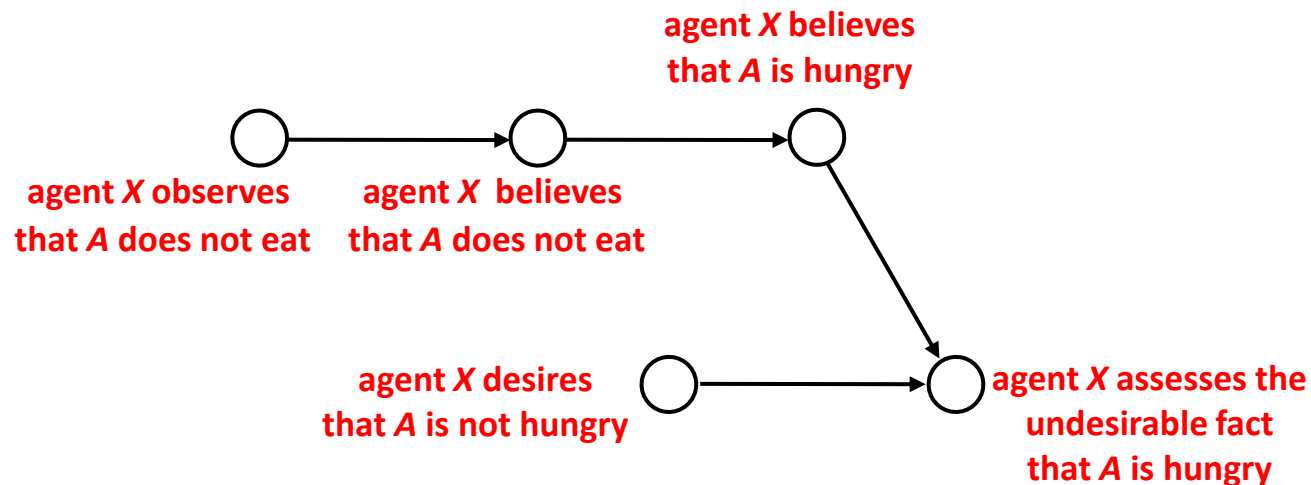
Dynamic relationships in the domain model



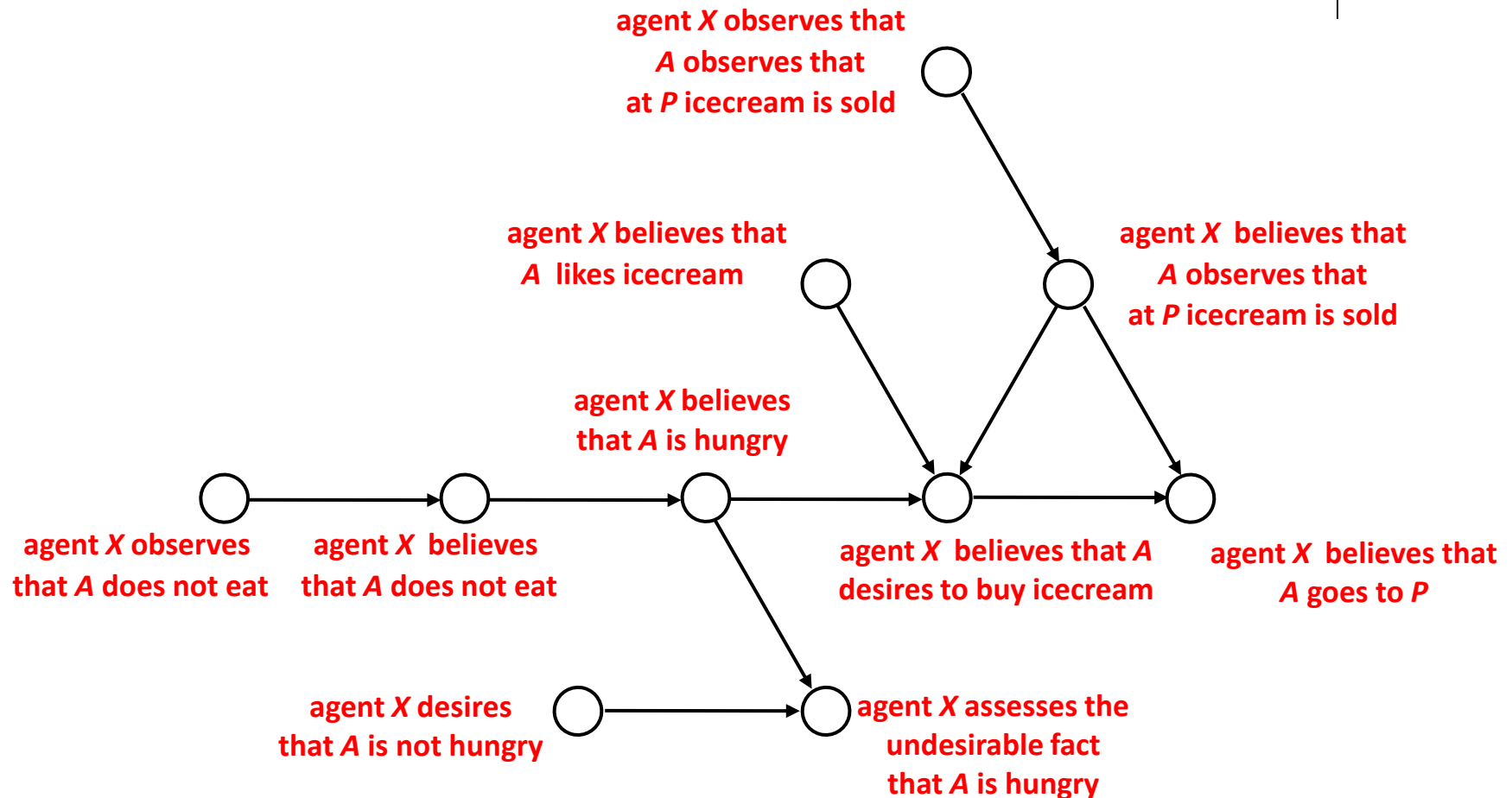
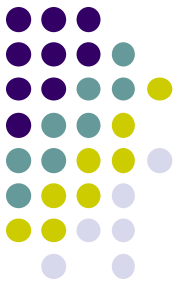
Dynamic relationships in the analysis model (forward)



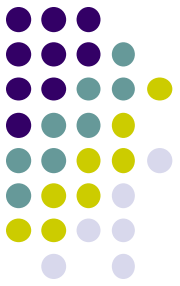
- Take perspective of intelligent agent
 - formulate **observations** and **beliefs** about Arnie
 - add a goal (desire)
 - add a conclusion (assessment)



Dynamic relationships in the analysis model (forward)



Steps from a domain model to analysis model (forward)



Embedding the concepts in a domain model relationship within beliefs or observations of the ambient agent; for example, the **domain model relationship**

E affects F

between concepts E and F

(for example, $E = 'A$ is hungry',

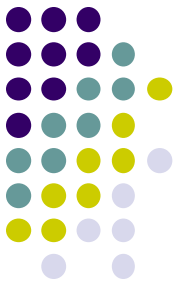
$F = 'A$ desires to buy icecream')

is **integrated** within the **analysis model** as:

ambient agent X believes that E affects

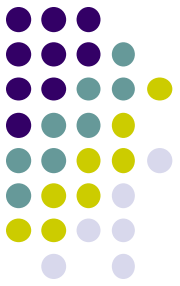
ambient agent X believes that F

Detailed analysis model: concepts



<i>concept</i>	<i>formalisation</i>
ambient agent X believes that A desires to buy icecream	belief(X: AGENT, desire(A: AGENT, performed(A, buy_icecream))))
ambient agent X believes that A observed that at P they sell icecream	belief(X: AGENT, observed(A: AGENT, icecream_at(P:POSITION))))
ambient agent X believes that A goes to P	belief(X, performed(A: AGENT, goto(P:POSITION))))
ambient agent X believes that A is hungry	belief(X: AGENT, hungry(A: AGENT))
ambient agent X believes that A likes icecream	belief(X: AGENT, likes(A: AGENT, icecream))
ambient agent X believes that A does not eat	belief(X: AGENT, not(performed(A: AGENT, eat))))
ambient agent X desires that A is not hungry	desire(X: AGENT, not(hungry(A:AGENT))))
ambient agent X assesses that the undesirable state occurs that A is hungry	assessment(X: AGENT, undesirable_state(hungry(A:AGENT))))

Analysis model: formalization of relations (forward)



ADR1 Generating beliefs from observations

If ambient agent X observes Y
then ambient agent X will believe Y

observed(X, Y) $\rightarrow$ belief(X, Y)

ADR2 Generating the belief that A is hungry

If ambient agent X believes that A does not eat
then ambient agent X will believe that A is hungry

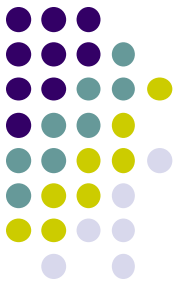
belief($X, \text{not}(\text{performed}(A, \text{eat}))) \rightarrow$ belief($X, \text{hungry}(A)$)

ADR3 Generating the belief that A desires to buy icecream

If ambient agent X believes that A is hungry
and ambient agent X believes that A observes that at P they sell icecream
and ambient agent X believes that A likes icecream
then ambient agent X will believe that A desires to buy icecream

**belief($X, \text{hungry}(A)$) &
belief($X, \text{observes}(A, \text{icecream_at}(P)))$ &
belief($X, \text{likes}(A, \text{icecream})$)
 $\rightarrow$ belief($X, \text{desire}(A, \text{performed}(A, \text{buy_icecream}))$)**

Analysis model: formalization of relations (forward)



ADR4 Generating the belief that A goes to the icecream

If ambient agent *X* believes that *A* desires to buy icecream
and ambient agent *X* believes that *A* observes that at *P* they sell icecream
then ambient agent *X* will believe that *A* will go to *P*

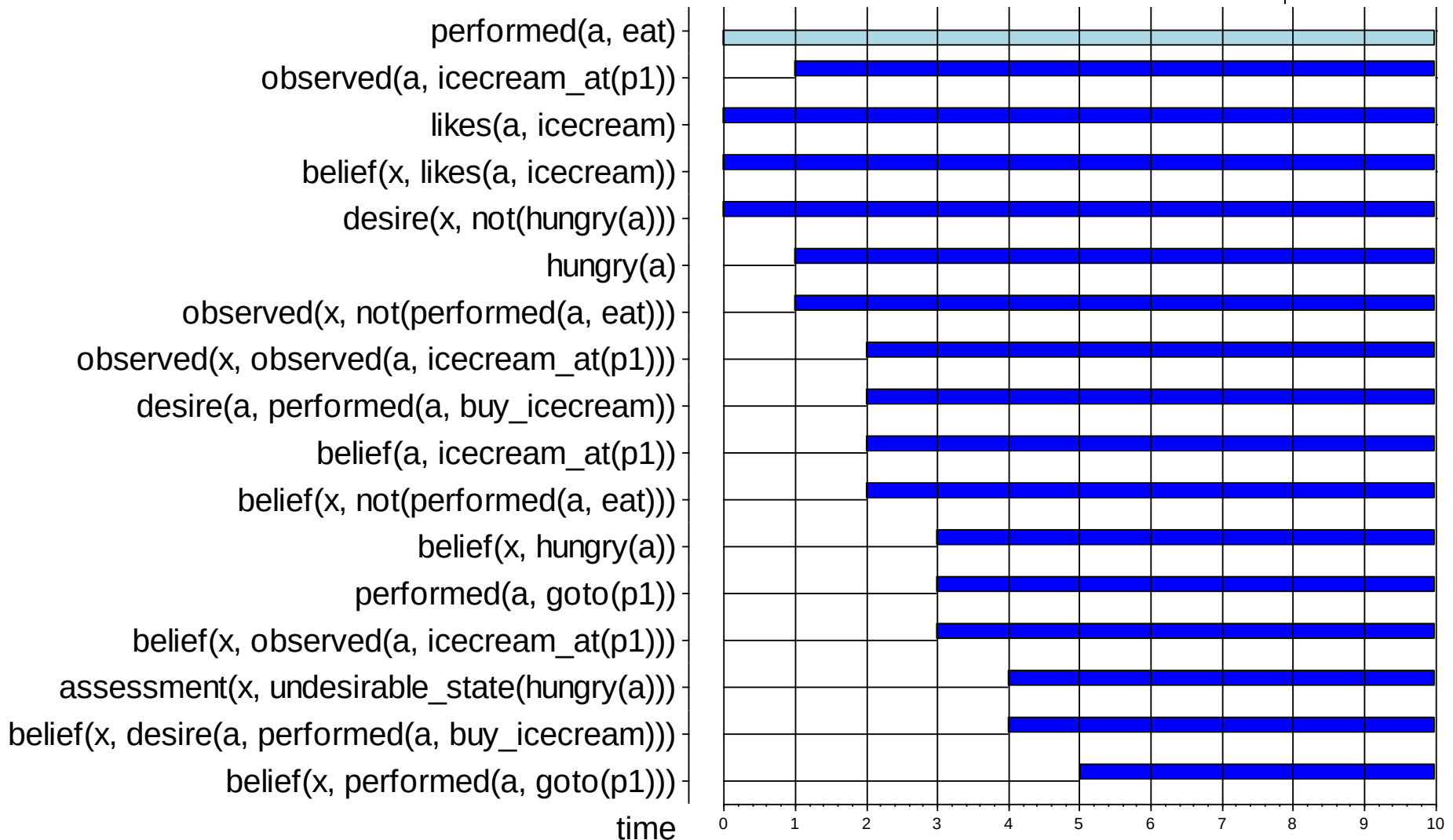
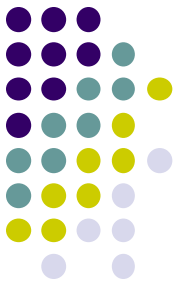
**belief(X, desire(A, performed(A, buy_icecream))) &
belief(X, observes(A, icecream_at(P)))
->> belief(X, performed(A, goto(P)))**

ADR5 Generating the assessment that A is in an undesirable state of being hungry

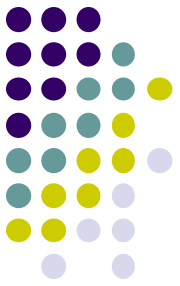
If ambient agent *X* believes that *A* is hungry
and ambient agent *X* desires that *A* is not hungry
then ambient agent *X* will assess that it is undesirable that *A* is hungry

**belief(X, hungry(A)) &
desire(X, not(hungry(A)))
->> assessment(X, undesirable_state(hungry(A)))**

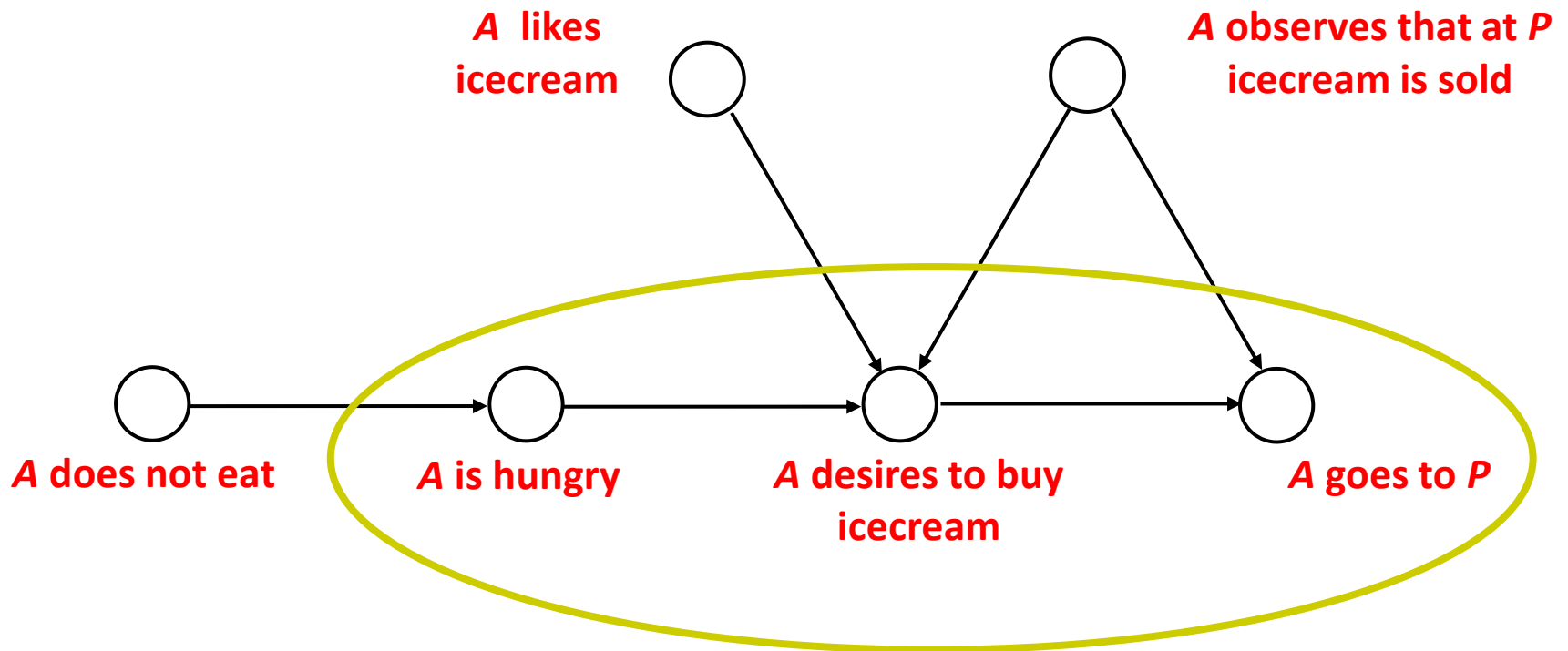
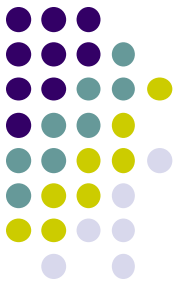
Example trace for the forward reasoning Analysis model



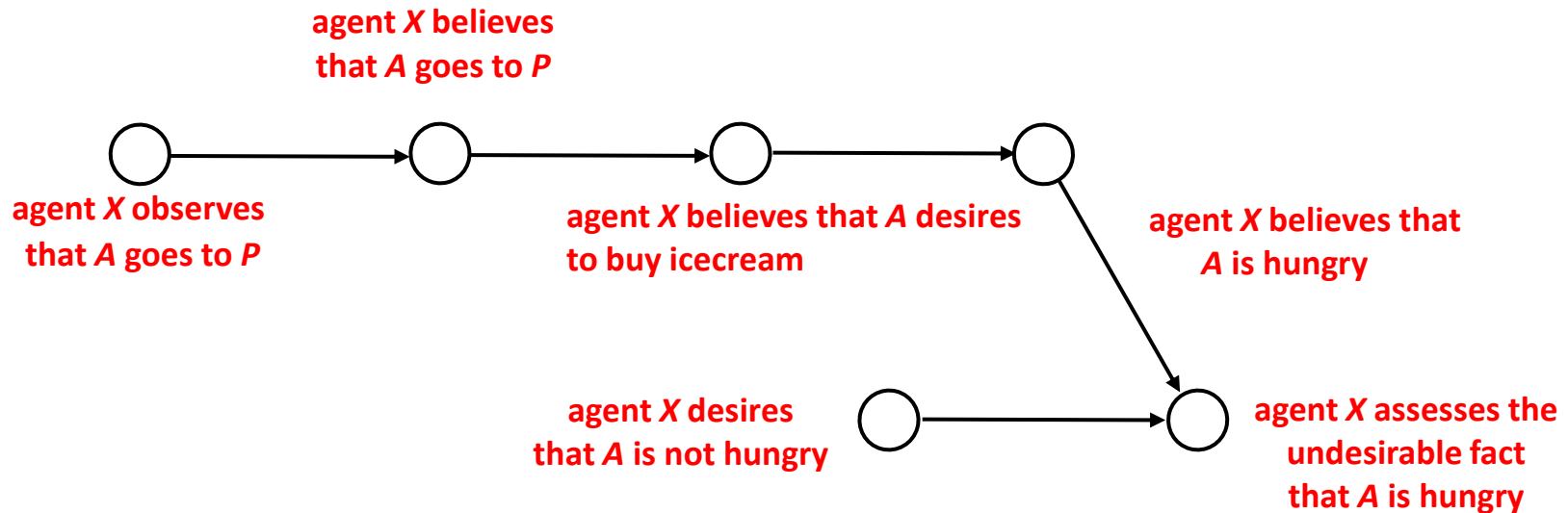
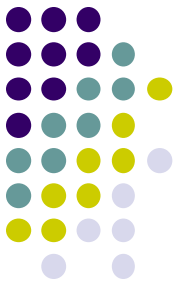
Analysis model via backward reasoning



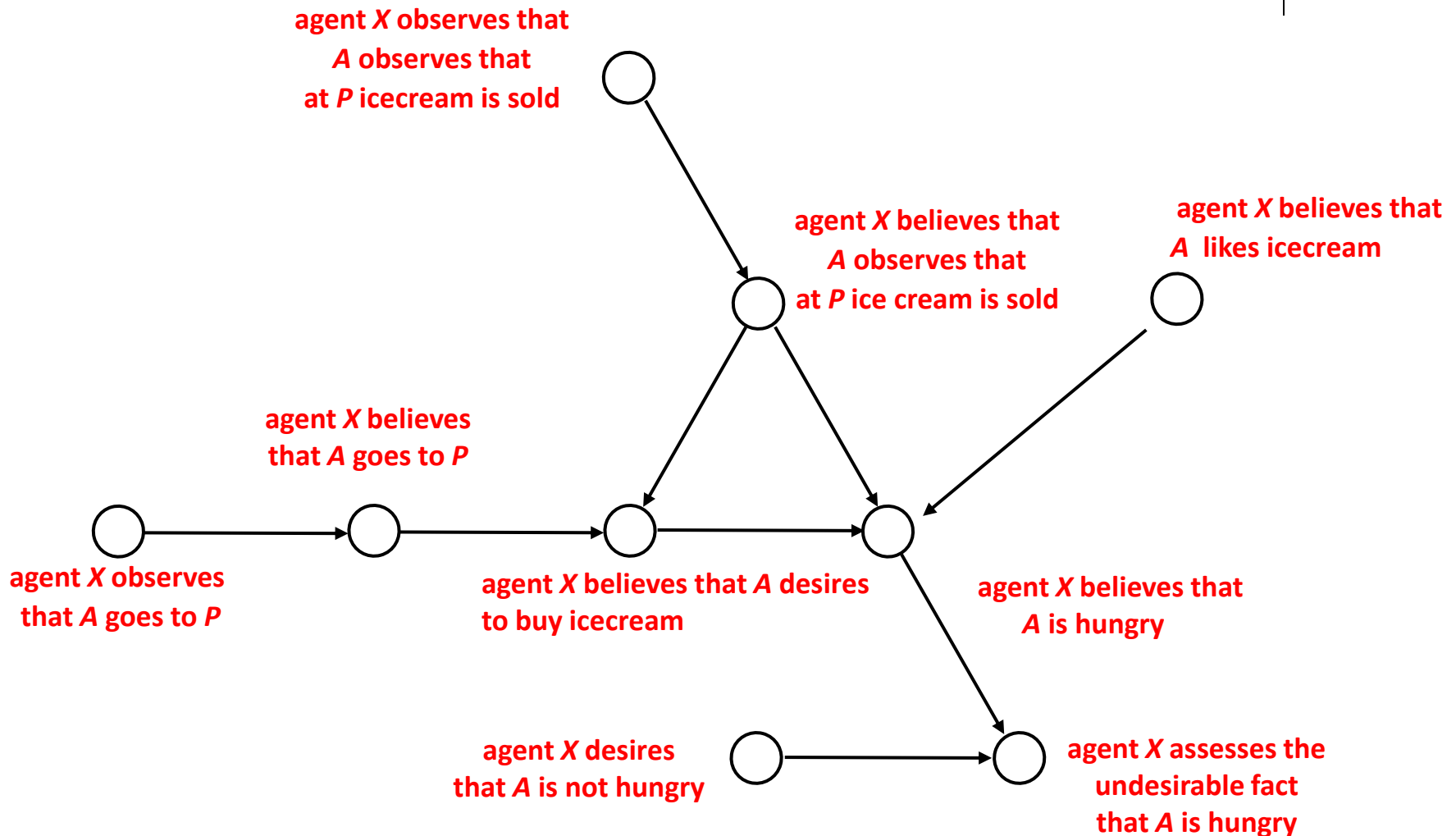
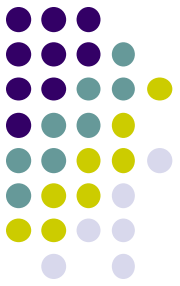
Same domain model



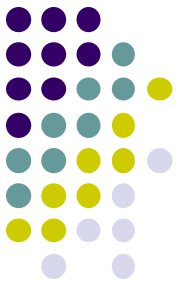
Dynamic relationships in the analysis model (backward)



Dynamic relationships in the analysis model (backward)



Analysis model: formalization of relations (backward)



ADR6 Abducing the belief that *A* desires to buy icecream

If ambient agent *X* believes that *A* goes to *P*
and ambient agent *X* believes that *A* observes that at *P* they sell icecream
and ambient agent *X* believes that *A* likes icecream
then ambient agent *X* will believe that *A* desires to buy icecream

belief(*X*, performed(goto(*P*))) &
belief(*X* observed(*A*, icecream_at(*P*))) &
belief(*X* likes(*A*, icecream))
->> belief(*X*, desire(*A*, performed(*A*, buy_icecream)))

ADR7 Abducing the belief that *A* is hungry

If ambient agent *X* believes that *A* desires to buy icecream
and ambient agent *X* believes that *A* observes that at *P* they sell icecream
then ambient agent *X* will believe that *A* is hungry

belief(*X*, desire(*A*, performed(*A*, buy_icecream))) &
belief(*X* observed(*A*, icecream_at(*P*)))
->> belief(*X*, hungry(*A*))

Steps from a domain model to analysis model (backward)



Embedding the concepts in a domain model relationship within beliefs of the ambient agent in a reverse manner, for example, the **domain relationship**

E affects F

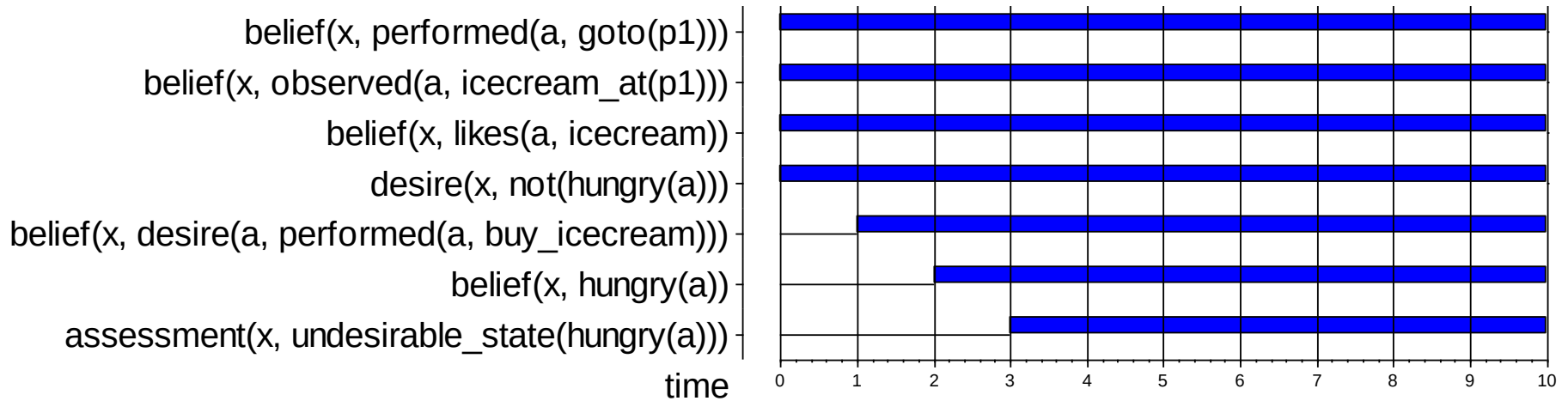
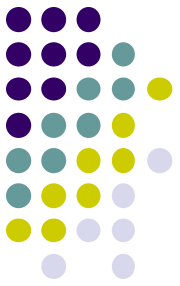
between concepts E and F

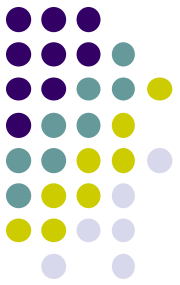
(for example, E = ' A desires to buy icecream',
 F = ' A goes to P ')

is **integrated** within the analysis model as the relationship between beliefs about them, and **reversing the direction** for two of the concepts:

ambient agent X believes that F affects
ambient agent X believes that E

Example trace for the backward reasoning Analysis model

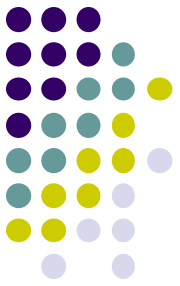




Overview of today's lecture

1. Integrating domain models within agent models
 - analysis model
 - support model
 - adaptation model
2. Translating a domain model into an analysis model
 - “ice cream” example of domain model
 - translation into analysis model
 - **generic process**

Generic process of developing analysis models



- Main purpose:
 - find out whether some **undesired state** occurs
- Main requirements:
 - a domain model
 - sensor information

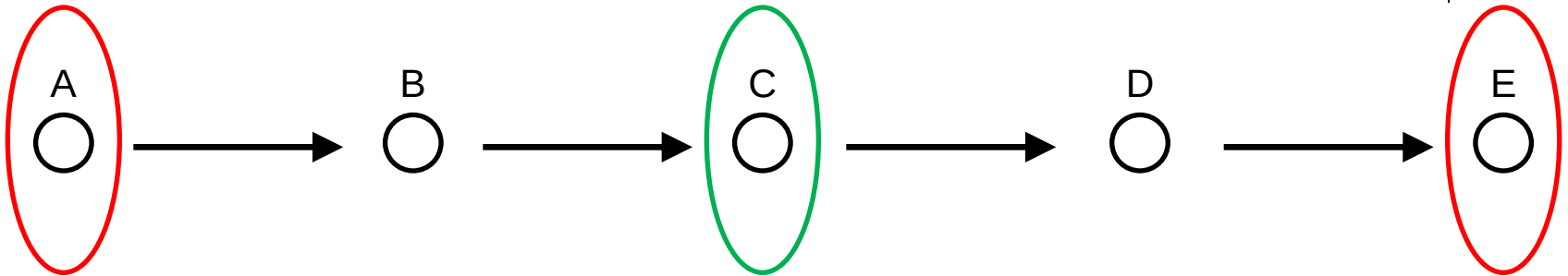
Domain Model → Analysis Model



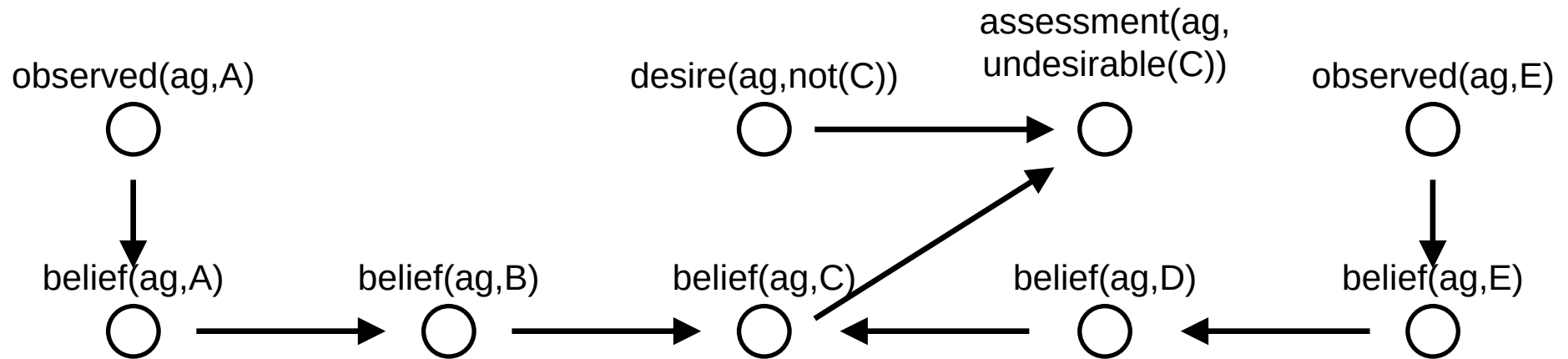
1. Decide which concepts of domain model can be **observed**
2. Introduce **observations** for these concepts
3. Decide which concept of domain model is (not) **desired**
4. Introduce a **desire** for this concept
5. Introduce **beliefs** for all concepts
6. Draw **arrows** from observations via beliefs to the desire (based on forward and/or backward reasoning)
7. Introduce an **assessment** for the concept from step 3
8. Draw **arrows** from the desire+belief to the assessment



Domain Model → Analysis Model

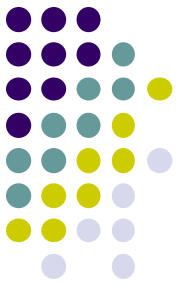


DOMAIN



ANALYSIS

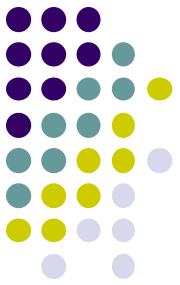
Additional Modelling Concepts



<i>concept</i>	<i>relation</i>
assessment (of an undesirable state)	assessment : AGENT x INFO_ELEMENT

Example:

- assessment(agent, undesirable_state(temp(25)))



Conclusion

- Method for integration in multiple steps:
 - **domain** model
 - formal description of human process
 - as developed during the first half of the course
 - **sensors**
 - used to measure certain states in the domain model
 - **analysis** model
 - uses domain model at run-time based on observations
 - matches *believed* human state to some *desired* state
 - ⇒ generates **assessments**